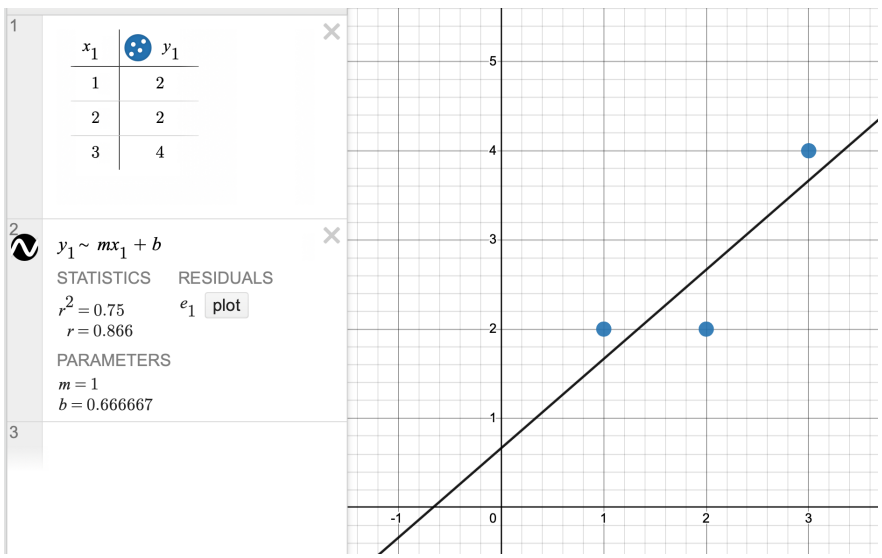
Regression - theory

Suppose we have three pairs of (x, y) values: (1,2), (2,2), (3,4). Open Desmos and plug them into a table, then make a linear model as we did before.



Desmos says that the linear regression model gives a line with a slope of 1 and a y-intercept of 2/3. We want to know how the calculation was done. It's actually pretty easy.

mean

First, we need the average or *mean* of the x and y values. One symbol for the mean is to put a little bar over the letter, and in talking about

them say "x-bar" and "y-bar".

$$\bar{x} = \frac{1+2+3}{3} = 2 \qquad \bar{y} = \frac{2+2+4}{3} = 8/3$$

using $\bar{x}$ and $\bar{y}$

Take each x value in the list of values and subtract the mean.

$$(x_1 - \bar{x}),\ (x_2 - \bar{x}),\ (x_3 - \bar{x})$$
$$= (1-2) + (2-2) + (3-2)$$
$$= -1,\ 0,\ 1$$

Do the same for y.

$$(y_1 - \bar{y}),\ (y_2 - \bar{y}),\ (y_3 - \bar{y})$$
$$(2 - 8/3),\ (2 - 8/3),\ (4 - 8/3)$$
$$-2/3,\ -2/3,\ 4/3$$

multiply, and add

Take the corresponding values from each resulting list and multiply them together. For example, the first term is $(x_1 - \bar{x}) \cdot (y_1 - \bar{y})$. (I'm using $\cdot$ as the symbol for multiplication).

$$-1 \cdot -2/3,\ 0 \cdot -2/3,\ 1 \cdot 4/3$$

Add to obtain the result

$$2/3 + 0 + 4/3 = 6/3 = 2$$

Th result of this calculation is often called $SSxy$. (Probably not because it's *so sexy*).

We're almost there!

Go back to the first list. Square each term. The first one is $(x_1 - \bar{x})^2$.

$$-1 \cdot -1, 0 \cdot 0, 1 \cdot 1$$

Add to obtain the second result.

$$1 + 0 + 1 = 2$$

This is often called $SSxx$.

equation

Divide the first result by the second ($SSxy/SSxx$). This is the slope of the line that gives the best fit by linear regression.

$$m = \frac{SSxy}{SSxx} = \frac{2}{2} = 1$$

To complete this analysis, we need the intercept, b.

Take advantage of the fact that the point corresponding to the means $(\bar{x}, \bar{y})$ is on the line. We won't prove this just yet.

The point $(2, 8/3)$ is on the line, so it satisfies the equation:

$$\bar{y} = m\bar{x} + b \qquad 8/3 = (1 \cdot 2) + b$$

Rearranging

$$b = \bar{y} - m\bar{x}$$
$$= 8/3 - (1 \cdot 2) = 2/3$$

And that's how the calculation is done.

Sigma notation

In statistics we try to simplify the notation. Consider a list of values:

$$x_1, x_2, \ldots x_n$$

When we want to talk about a value from the list, without specifying exactly which one, call it x_i, and write the sum of all the values using the Greek letter capital sigma (Σ) as

$$\sum x_i$$

where

$$\sum x_i = x_1 + x_2 \cdots + x_n$$

If we wanted to be really precise, we might write

$$\sum_{i=1}^{i=n} x_i = x_1 + x_2 \cdots + x_n$$

The i stands for *index*.

Using this notation the definition of the mean is

$$\bar{x} = \frac{1}{n} \sum x_i = \frac{1}{n}(x_1 + x_2 \cdots + x_n)$$

Variance

For a given set of values, we are often interested in how closely clustered they are. How different are the values from the mean, in aggregate?

We could subtract the mean from each value, as we did above

$$\sum (x_i - \bar{x}) = (x_1 - \bar{x}) + (x_2 - \bar{x}) \cdots + (x_n - \bar{x})$$

There is a problem, however. Unless the values are all the same, some are larger and some smaller than the mean, so some of the terms in the sum are negative and some positive, and there will be cancellations.

In fact, the expression above always evaluates to zero. Can you rearrange terms to prove it?

To make all the differences positive or at least non-negative, it is useful to square them

$$\sum (x_i - \bar{x})^2 = (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 \cdots + (x_n - \bar{x})^2$$

This is just what we called $SSxx$ before.

There is one more manipulation. Divide the whole thing by the number of items, n, and give it a special name, the *variance* of x. Write V for variance:

$$V_x = \frac{1}{n} \sum (x_i - \bar{x})^2$$

When there are two variables, x and y, we can also compute the *covariance*, C:

$$C_{xy} = \frac{1}{n} \sum (x_i - \bar{x}) \cdot (y_i - \bar{y})$$

If we divide one by the other, the n's cancel and we have

$$\frac{C_{xy}}{V_x} = \frac{SSxy}{SSxx} = m$$

where m is the slope of the linear regression equation. And as before, $b = \bar{y} - m\bar{x}$.

another method

There is a second formula to compute the slope and intercept of the linear regression equation. We will show later that the two forms are actually the same statement in different terms.

Compute $x \cdot y$ for each position and then find the mean of those values.

$$\frac{1 \cdot 2 + 2 \cdot 2 + 3 \cdot 4}{3} = 6$$

Subtract $\bar{x} \cdot \bar{y}$

$$6 - 2 \cdot 8/3 = 2/3$$

Similarly, compute $x \cdot x$ for each position and find the mean of those values.

$$\frac{1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3}{3} = \frac{14}{3} = 4 + 2/3$$

Subtract $\bar{x} \cdot \bar{x}$

$$14/3 - 4 = 2/3$$

Divide the first result by the second to obtain $m = 1$.

derivation of the formulas

Suppose we have three points (x_1, y_1), (x_2, y_2) and (x_3, y_3). They nearly lie on one line, but not exactly.

We want to find a linear regression equation $y = Ax + B$, where A and B are the slope and intercept that give the "best fit" to the data, defined in the following way.

The error of the predicted value for y from each x_i is different than what is actually in the data by

$$e_i = y_i - (Ax_i + B)$$

The error squared is

$$e_i^2 = \ [\ y_i - (Ax_i + B)\ ]^2$$

We want the total of all the squared errors to be a minimum. Note that subtracting y_i from $Ax_i + B$ gives the same result for the square.

That's because

$$(c-d)^2 = \ [\ (-1)(d-c)\ ]^2 \ = (-1)^2(d-c)^2 = (d-c)^2$$

The total is

$$e_{total} = \sum_{i=1}^{i=n} [\ y_i - (Ax_i + B)\ ]^2$$

The x and y values are *constants* for a given problem, but A and B are variables. We can take the partial derivatives of e with respect to A and B. With taking a partial derivative, say $\partial e/\partial A$, the other variable B is treated as a constant.

The sum does not cause any difficulty, since the derivative of a sum is the sum of the derivatives.

Thus (suppressing the index)

$$\frac{\partial e}{\partial A} = \sum [\ y_i - (Ax_i + B)\ ]\ (2)(-x_i)$$

$$= 2\sum (Ax_i^2 - x_i y_i + Bx_i)$$

That factor of $-x_i$ comes from the chain rule applied to the term $-(Ax_i + B)$.

The second one is the same except for that factor of x_i.

$$\frac{\partial e}{\partial B} = 2\sum [\ y_i - (Ax_i + B)\ ](-1)$$

$$= 2\sum [\ Ax_i - y_i + B\ ]$$

We set both of these partial derivatives equal to zero to find a minimum, and write the individual sums. The factors of 2 go away and we have

$$A\sum x_i^2 - \sum x_i y_i + B\sum x_i = 0$$

$$A\sum x_i - \sum y_i + \sum B = 0$$

Note that $\sum B = NB$. We could use some linear algebra (like Cramer's rule) but instead, easily solve the second one for B

$$NB = -A\sum x_i + \sum y_i$$

Substitute back into the first equation:

$$A\sum x_i^2 - \sum x_i y_i - \frac{1}{N}A\sum x_i \sum x_i + \frac{1}{N}\sum x_i \sum y_i = 0$$

Gather terms, rearrange, and divide by the cofactors of A

$$A = \frac{\sum x_i y_i - (1/N)\sum x_i \sum y_i}{\sum x_i^2 - (1/N)\sum x_i \sum x_i}$$

If we divide the top and bottom by $1/N$ we obtain just the right number of N's to turn all those sums into means.

$$= \frac{\mu(x_i y_i) - \mu(x)\mu(y)}{\mu(x_i x_i) - \mu(x)\mu(x)}$$

connection to a second formula

There is a second formula for A, namely the one we used in the first example

$$A = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

We want to show that these two are equivalent. Start with the denominator of this formula, using m for $\bar{x}$ and suppress the index i. We have

$$\sum(x - m)^2$$

Expand the square

$$= \sum x^2 - \sum 2mx + \sum m^2$$

Since m is a constant

$$= \sum x^2 - 2m \sum x + nm^2$$

But $(\sum x)/n = m$ so $\sum x = nm$ and then

$$= \sum x^2 - nm^2$$

$$= \sum x^2 - m \sum x$$

If at the end we multiply both the numerator and denominator by $1/n$ and use μ for the mean:

$$1/n \cdot \sum (x - m)^2 = \frac{1}{n}(\sum x^2 - m \sum x)$$

$$= \mu(x^2) - \mu(x)\mu(x)$$

This matches the denominator of the previous formula.

For the numerator:

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum (x_i y_i - x_i \bar{y} - y_i \bar{x} + \bar{x}\bar{y})$$

$$= \sum x_i y_i - \sum x_i \bar{y} - \sum y_i \bar{x} + \sum \bar{x}\bar{y}$$

$$= \sum x_i y_i - \bar{y} \sum x_i - \bar{x} \sum y_i + n\bar{x}\bar{y}$$

Multiplying by $1/n$ we have then

$$= \mu(x_i y_i) - \bar{y}\bar{x} - \bar{x}\bar{y} + \bar{x}\bar{y}$$

$$= \mu(x_i y_i) - \bar{x}\bar{y}$$

So the whole thing is a match.

solving for B

Recall that

$$NB = -A\sum x_i + \sum y_i$$
$$B = -A\bar{x} + \bar{y}$$
$$A\bar{x} + B = \bar{y}$$

This not only gives a way to calculate B but shows that $(\bar{x}, \bar{y})$ is on the line.